

Lab Assignment: Case Study on Class Modeling

Useful References -

1. What is a class diagram? - [Class Diagrams](#)
2. [OO-Relationships](#)

Tools - [StarUML](#) , [draw.io](#), [SmartDraw](#)

Prepare a detailed class diagram including all classes, their methods and attributes (with visibility notation), relevant relationships (dependencies, generalizations, associations) with multiplicities, enumeration, etc. as described by the case study given below.

Smart Water Distribution Management System

A city authority plans to deploy a smart system to monitor and control water distribution across residential and commercial zones. The overall system is operated under the supervision of city officials who are responsible for policy decisions, tariff management, and performance monitoring, while consumers and maintenance staff interact with the system for daily operational activities.

Each individual interacting with the system maintains a digital identity containing personal and contact details, along with access permissions based on their assigned responsibilities within the city's water authority framework. Users must be able to authenticate themselves, update profile information, and access functionalities relevant to their roles.

The city is divided into multiple distribution zones, each managed as part of the city's water infrastructure. Every zone contains several water meters installed at consumer locations. These meters continuously record water consumption and can be operational, faulty, or under maintenance. Meter readings are collected periodically and stored for monitoring and billing purposes.

Consumers are billed automatically based on recorded consumption and applicable tariff slabs defined by the city authority. Bills are generated at regular intervals, and consumers can make digital payments and review their billing and payment history. Unpaid or overdue bills must be identified by the system.

Maintenance staff are assigned tasks related to faulty meters or pipeline inspections within city zones. Once an issue is resolved, the system updates the relevant records. The system also sends alerts related to bill generation, maintenance schedules, service disruptions, or abnormal consumption. City officials can generate analytical reports and update tariff policies to ensure efficient water distribution.